TAMPER
STATION

TAMPER STATION BUILD GUIDE

Version 09.09.2025

PROLOG

Before you begin on your journey, a word of caution.

Please, read the **entire manual** before you start assembly.

GOOD LUCK AND GOOD COFFEE ANYTIME!

TABLE OF CONTENTS

INTRODUCTION

PART PRINTING GUIDELINES
We recommend you follow these Guidelines.

3D PRINTING PROCESS:
Fused Deposition Modeling (FDM)

MATERIAL:
PLA

LAYER HEIGHT:
Recommended: 0.2mm

EXTRUSION WIDTH:
Recommended: Forced 0.4mm

INFILL TYPE:
Gyroid

INFILL PERCENTAGE:
Recommended: 40%

WALL COUNT:
Recommended: 4

SOLID TOP/BOTTOM LAYERS:
Recommended: 5

FILE NAMING
By this time, you should have already downloaded our STL/3MF files. You might have noticed that we have used a unique naming convention for the files. This is how to use them.

PRIMARY COLOR:
Example: BaseBottom.3mf
These files will have nothing at the start of the filename.

ACCENT COLOR:
Example: [a]_58mmTamper.3mf
We have added “[a]” to the front of any STL file that is intended to be printed with accent color.

QUANTITY REQUIRED:
Example: [a]_BaseSide_Clean_x2.3mf
If any file ends with “_x#”, that is telling you the quantity of that part required to build the machine.

SUPPORT AND PRINT ORIENTATION:
Most files will print **without support**, if your 3D printer is well calibrated!

Except for one file of the base if you want it to look extra clean. The "support structure above print bed" must be activated there! See chapter “Print Orientation”

OVERVIEW OF TOOLS

TOOLS YOU WILL NEED

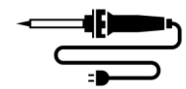
You will need the following tools. You should also be familiar with them and know how to use them.

SCREWDRIVER

SCALPEL

SUPER-GLUE

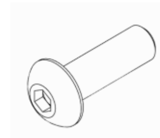
SOLDERING IRON



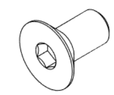
STRING CUTTER

PARTS LIST

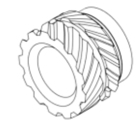
You can source your parts from Amazon, AliExpress, eBay or from your local parts store.



BHCS



FHCS



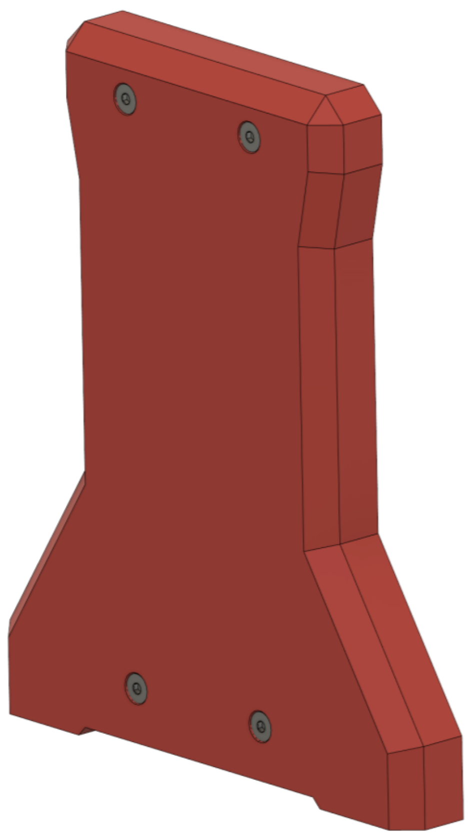
Heat set insert

Part	Properties	Quantity	Description
M3x10 FHCS	Countersunk head M3 with 10mm length	20	Mounting the tamper (2x), tamper rack (2x), base top (4x), base bottom (4x), base sides (8x)
M3 Heat set insert	Heat set insert M3 OD 4.5mm with 6mm length	18	Mounting the tamper rack (2x), base top (4x), base bottom (4x), base sides (8x)
M3x35 BHCS	Lens head M3 with 35mm length	1	Mounting the axle
M3x4 BHCS	Lens head M3 with 4mm length	1	Securing the handle rod
OD 8mm Aluminium rod	Aluminium rod with 200mm length and OD 8mm	1	Lever handle rod
16x25x1.8 Compression spring	Compression spring OD 16mm with 25mm length and 1.8mm wire diameter	1	Main spring for the right contact pressure
OD 8mm Rubber feet	Rubber feet with OD 8mm and thickness 2mm	4	Against slipping when tamping

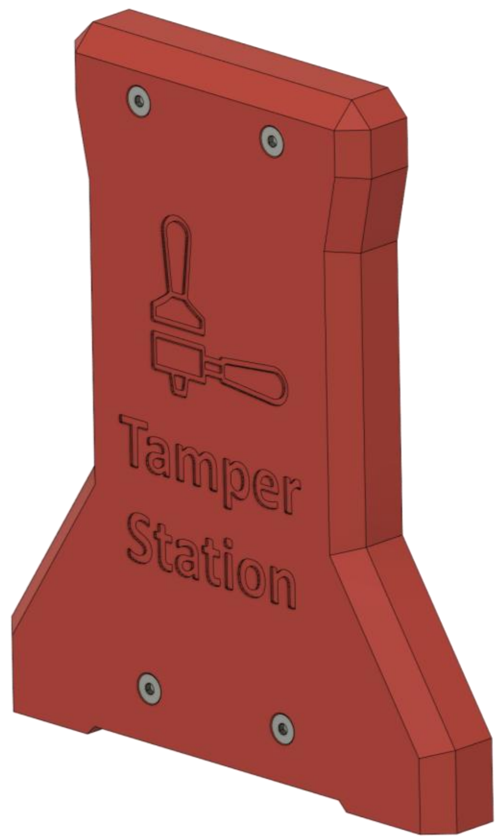
CHOOSE YOUR DESIGN

CHOOSE YOUR DESIGN

You have different designs to choose from for your 3D print. Print two side plates (_x2) of your choice.



[a]_BaseSide_**Clean**_x2.3mf

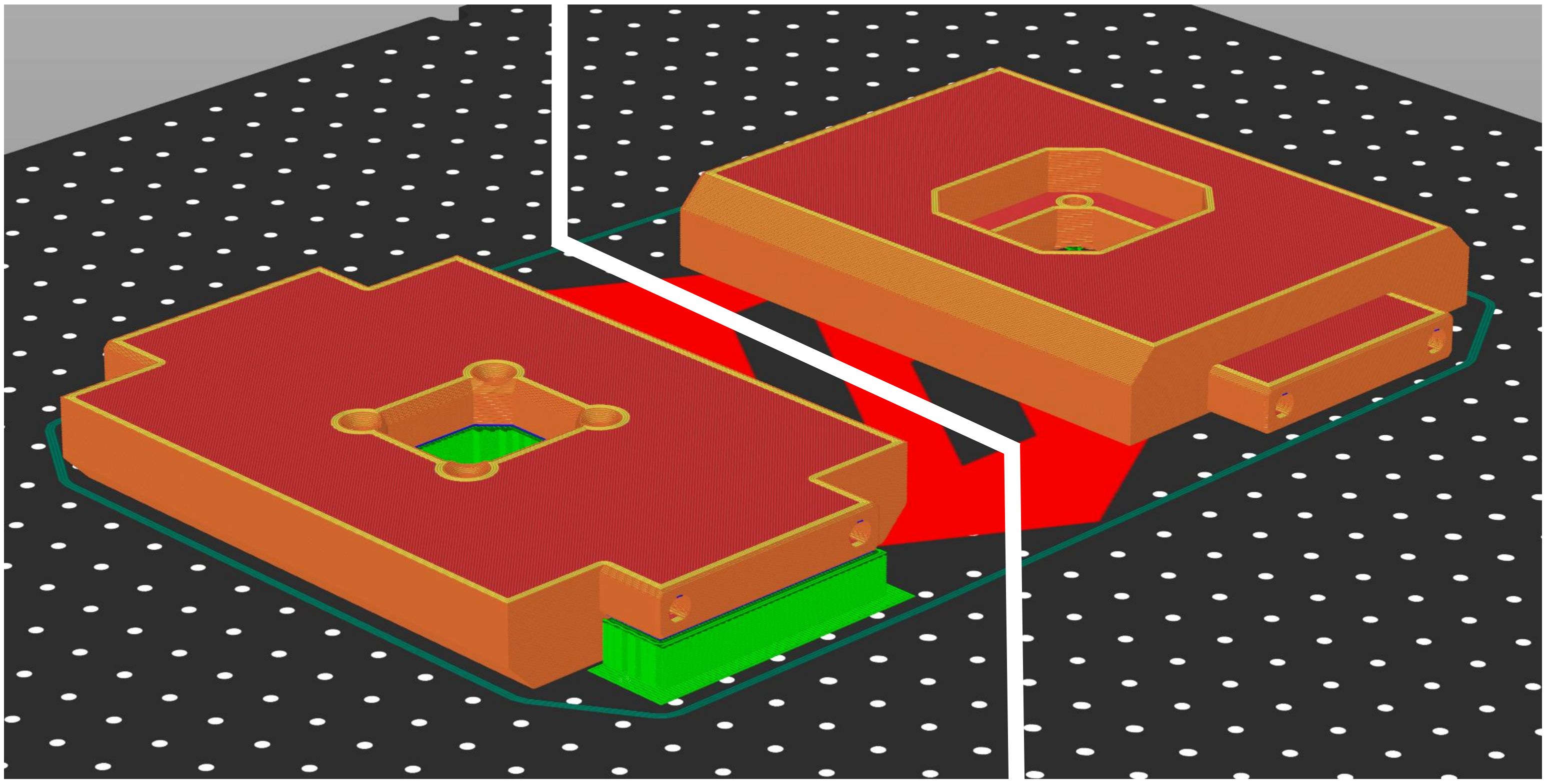


[a]_BaseSide_**WithText**_x2.3mf

PRINT ORIENTATION

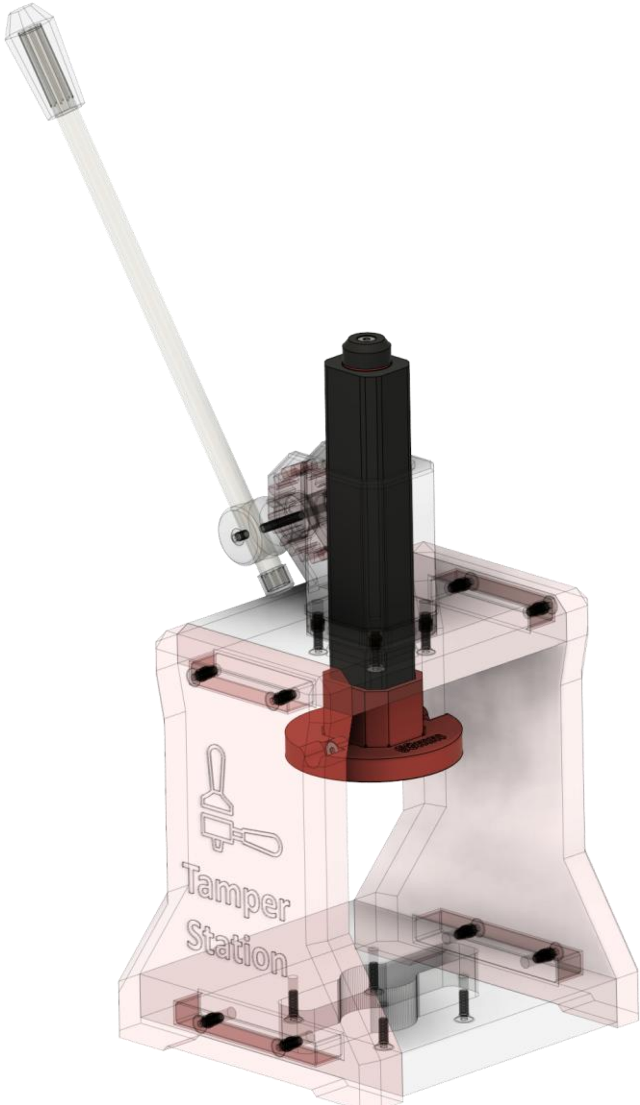
PRINT ORIENTATION

All parts are designed to print without support when properly oriented. The parts in the .3mf files are already correctly oriented. Nevertheless, if you want a perfect finish on the outside of your tamper station, you can print the BaseTop.3mf file upside down with support.



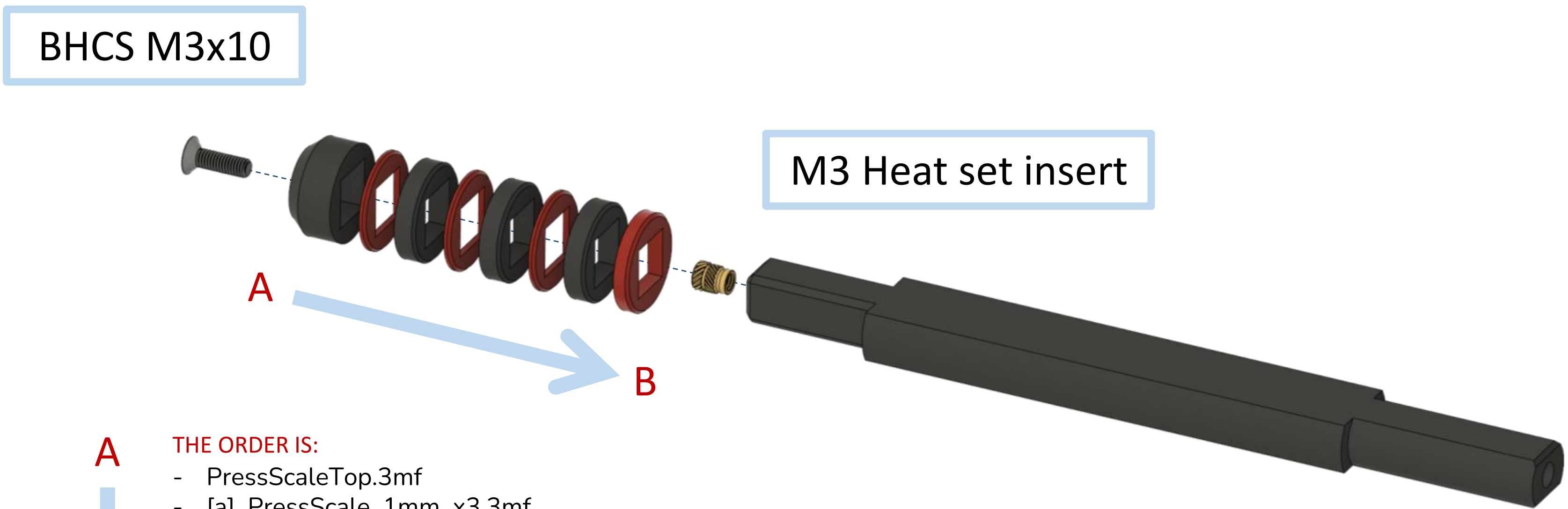
Print orientation for best outside finish (with support)

Print orientation without support



TAMPER RACK ASSEMBLY

TAMPER RACK



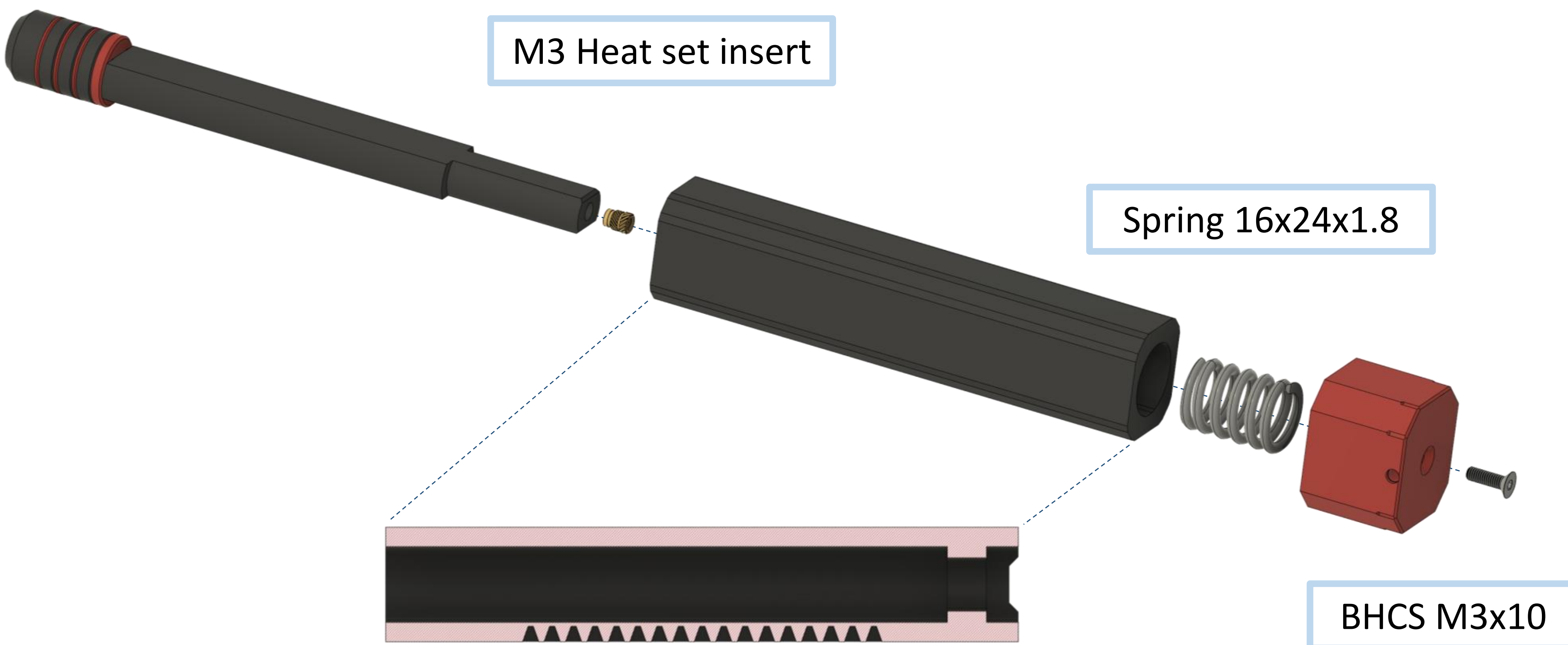
A

THE ORDER IS:
- PressScaleTop.3mf
- [a]_PressScale_1mm_x3.3mf
- PressScale_3.5mm_x3.3mf
- [a]_PressScale_1mm_x3.3mf
- PressScale_3.5mm_x3.3mf
- [a]_PressScale_1mm_x3.3mf
- PressScale_3.5mm_x3.3mf
- PressScaleBottom.3mf

B

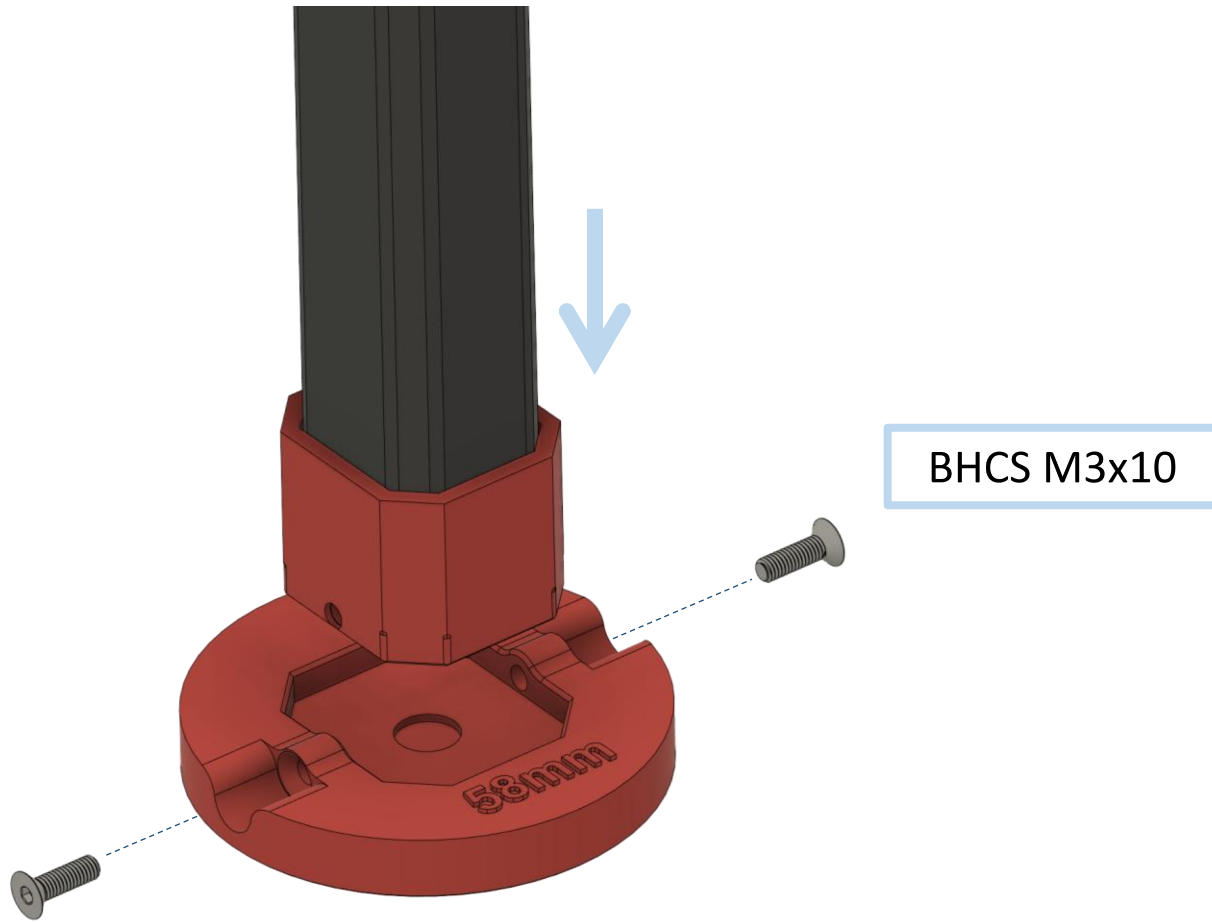
If you are using a different spring, look at the “**Calibration**” chapter.

TAMPER RACK

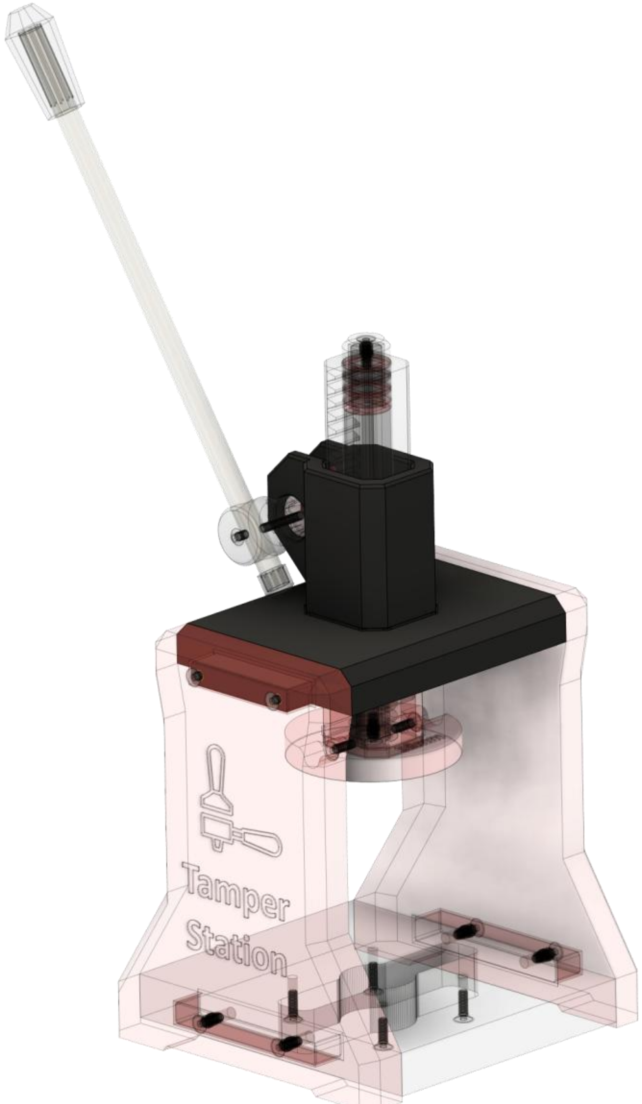


TIP:
Check that the housing is correctly oriented.

TAMPER RACK

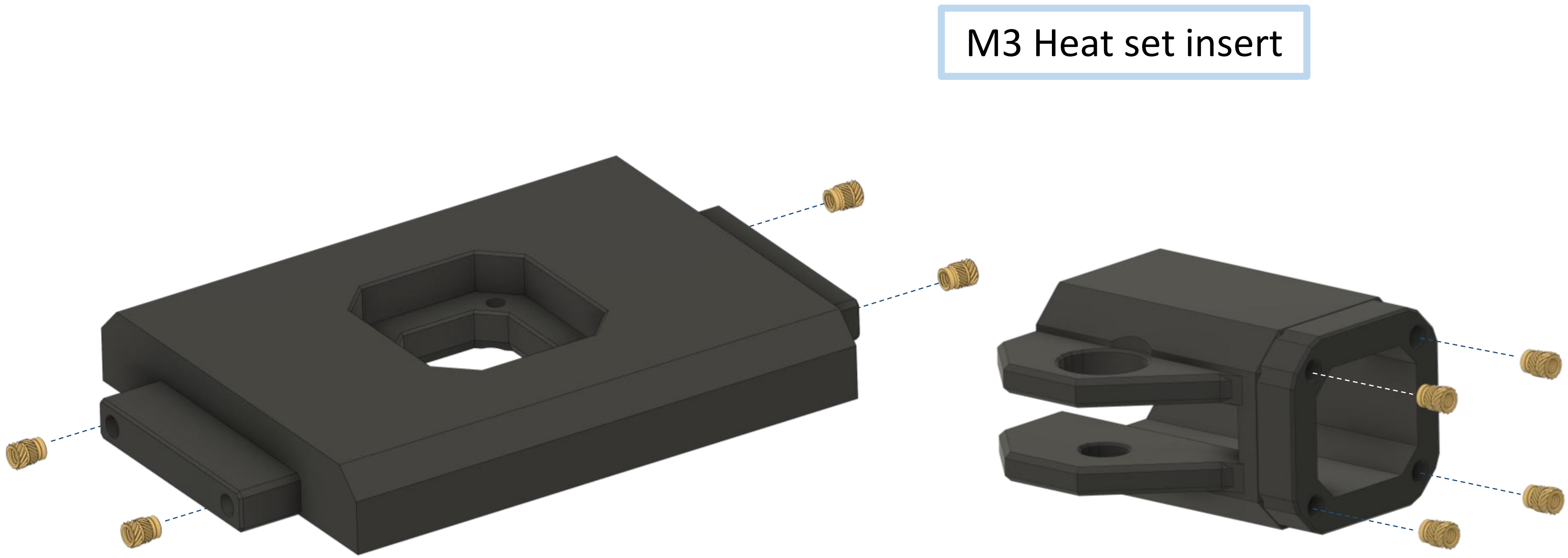


TIP:
Be careful not to overtighten the screws, as they are screws that are screwed directly into the plastic.

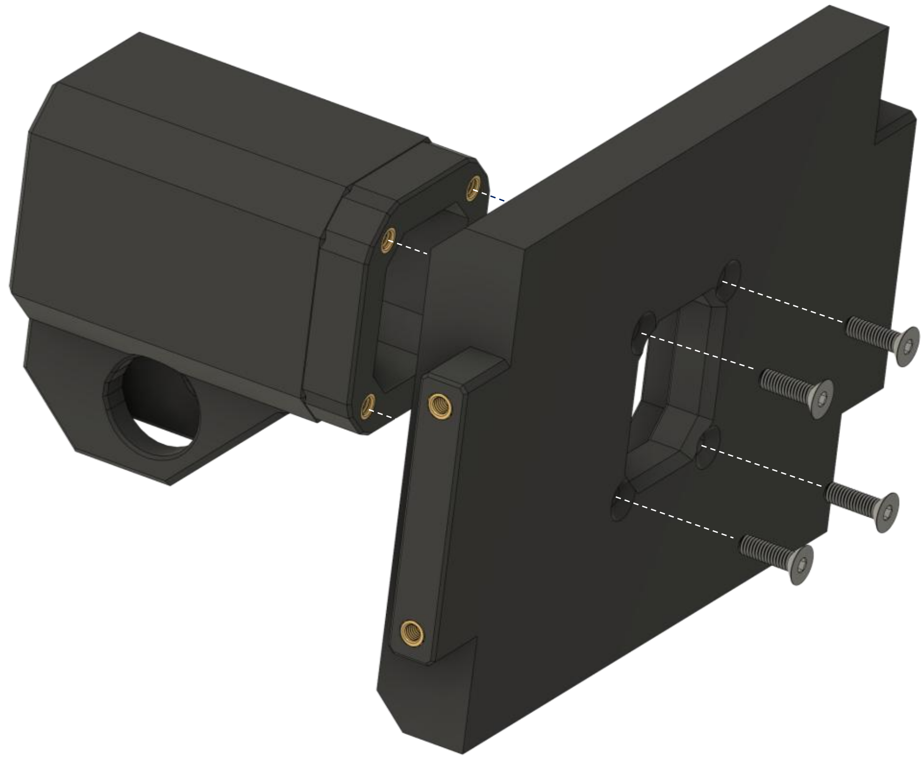


BASE TOP ASSEMBLY

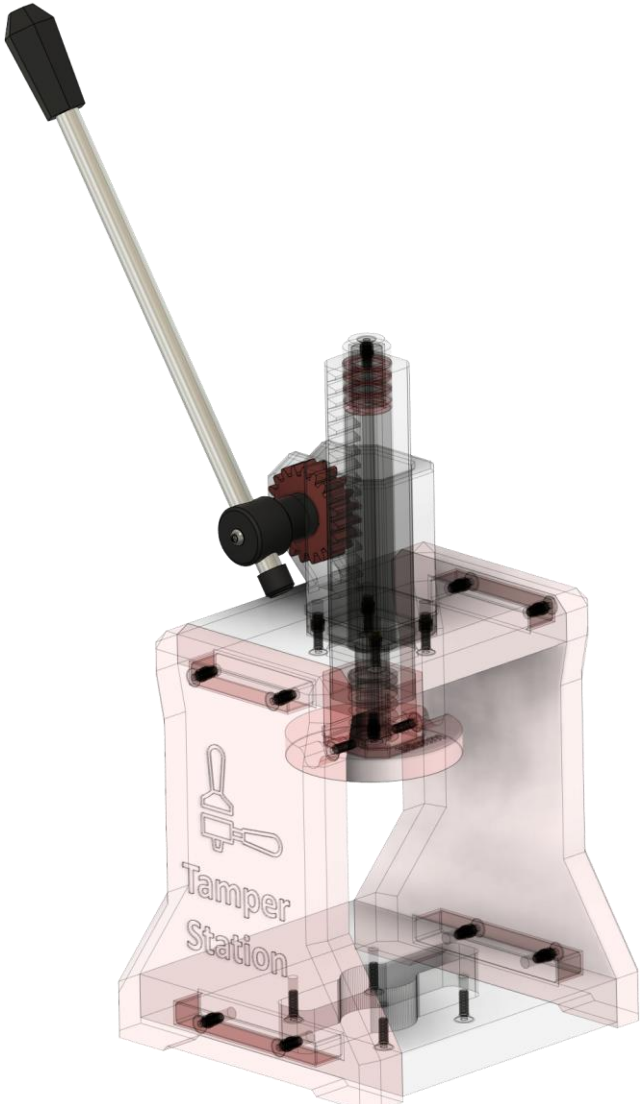
BASE TOP



BASE TOP

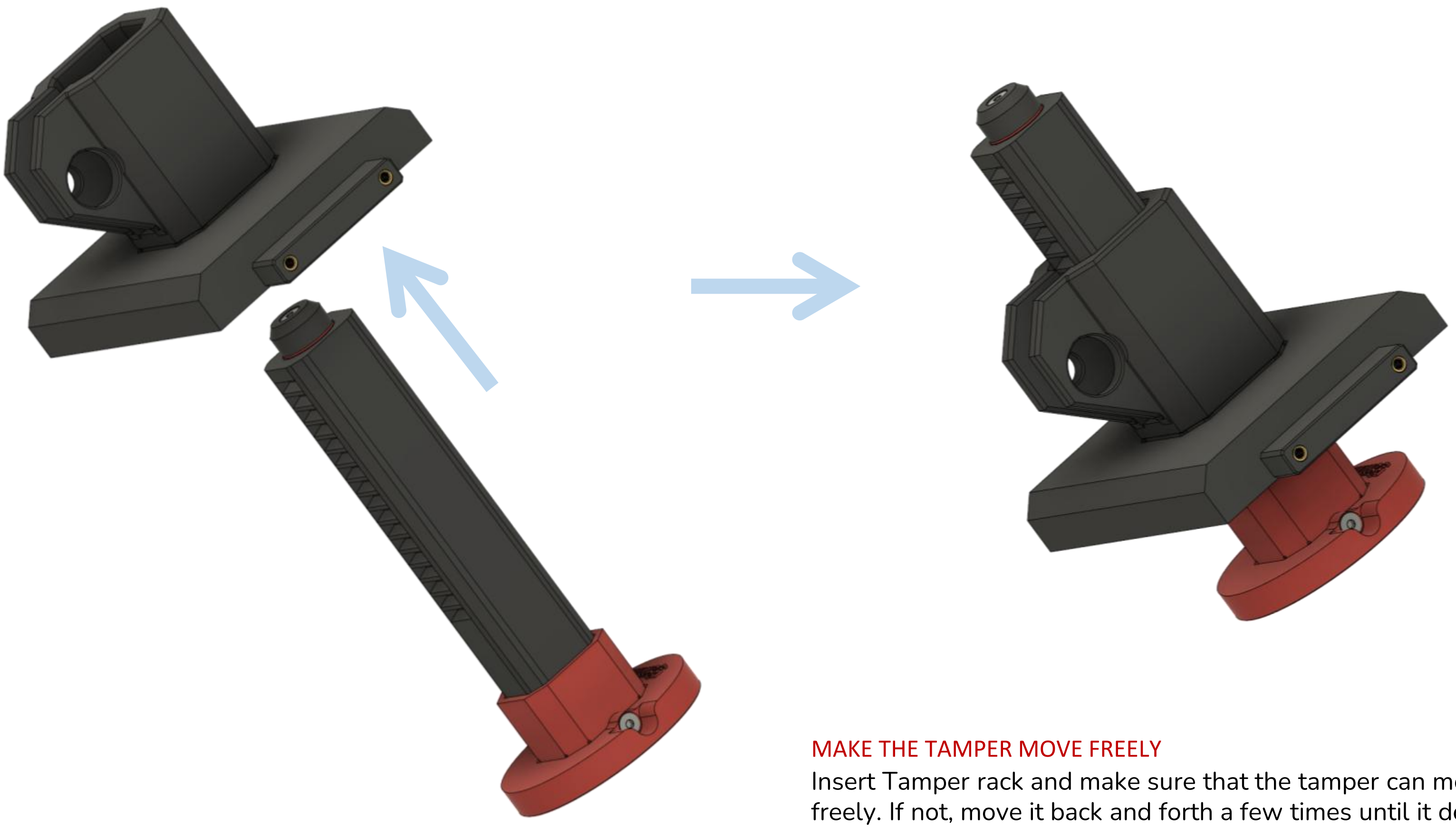


BHCS M3x10



MECHANICS ASSEMBLY

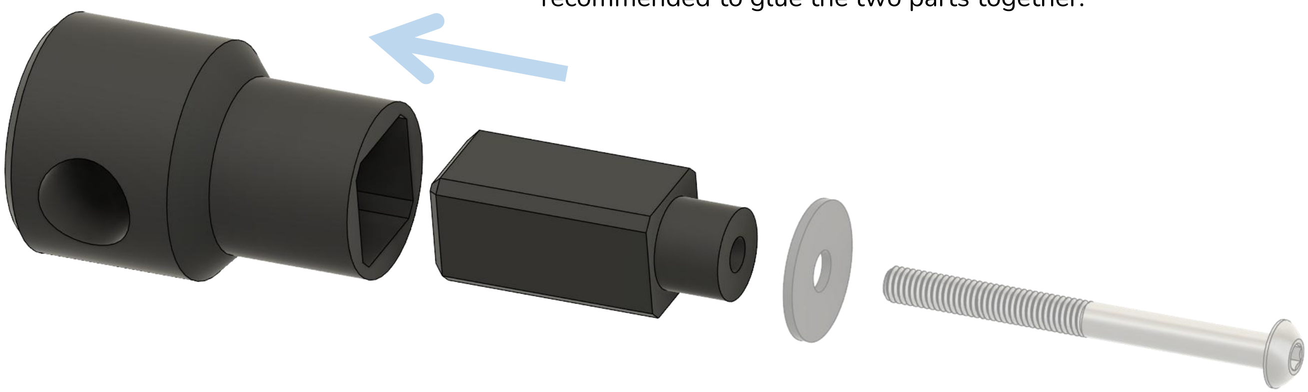
MECHANICS



MAKE THE TAMPER MOVE FREELY

Insert Tamper rack and make sure that the tamper can move freely. If not, move it back and forth a few times until it does.

MECHANICS

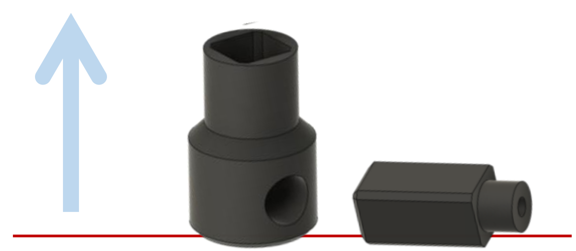


GLUE AXLE

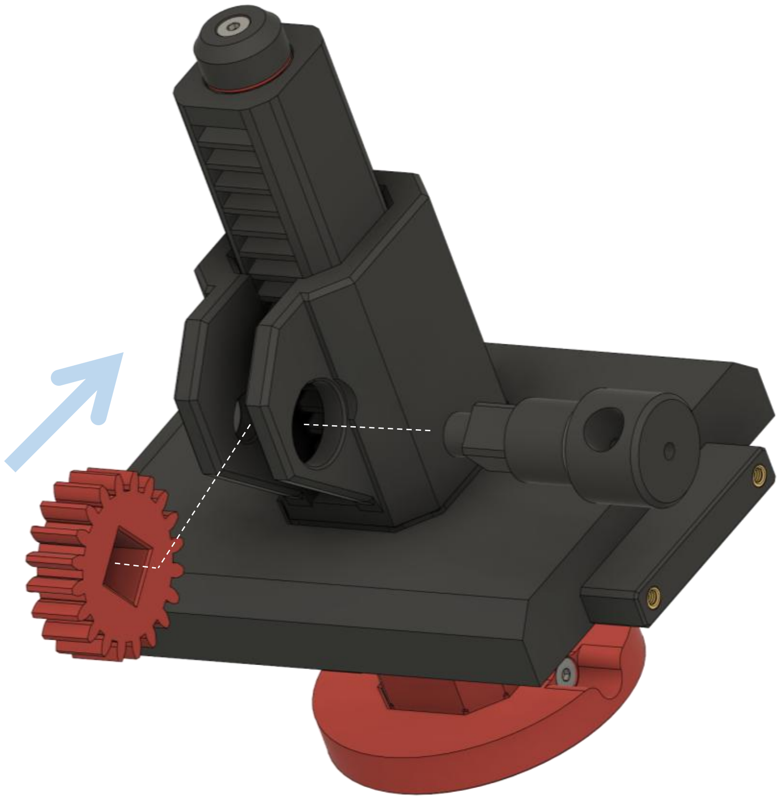
Glue the axle together using **superglue**. The axle can be used without gluing it, but for maximum stability, it is recommended to glue the two parts together.

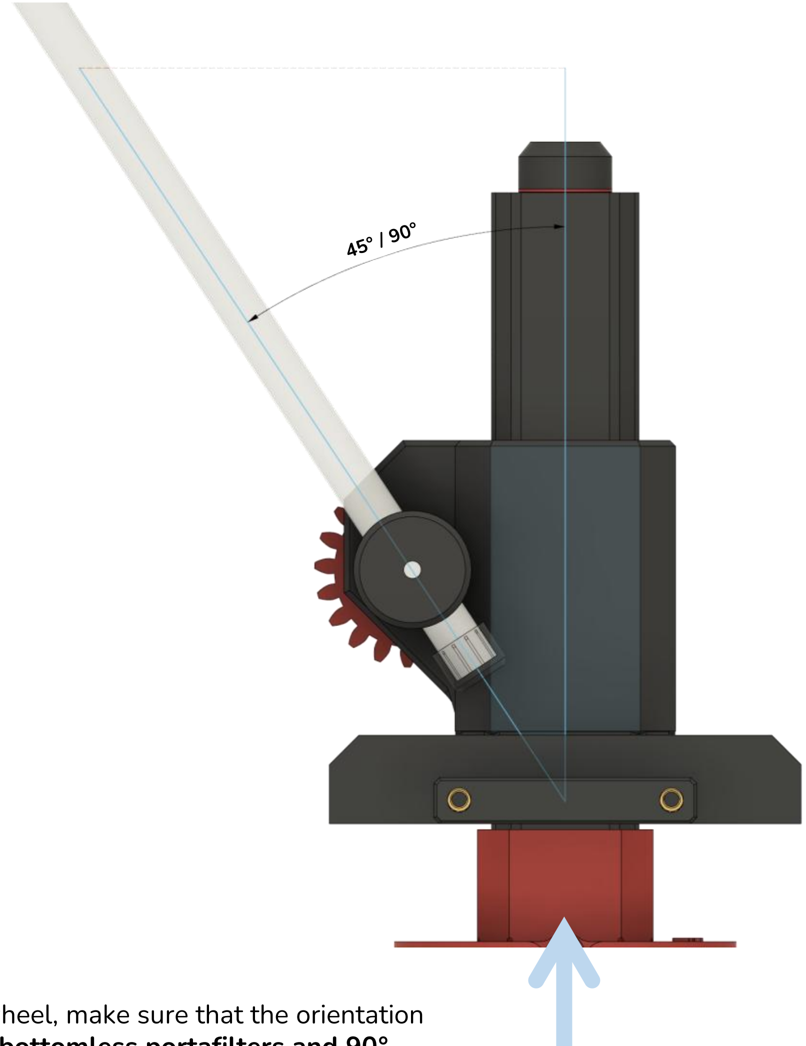
PRINT DIRECTION

For maximum stability, it is recommended to print these two parts in the direction shown.



MECHANICS





MOUNTING THE GEAR:

When you mount the gear wheel, make sure that the orientation of the lever is **about 45° for bottomless portafilters and 90° normal portafilters** to the rear when the tamper rack is fully retracted. This depends on your "Bottom Base" selection.

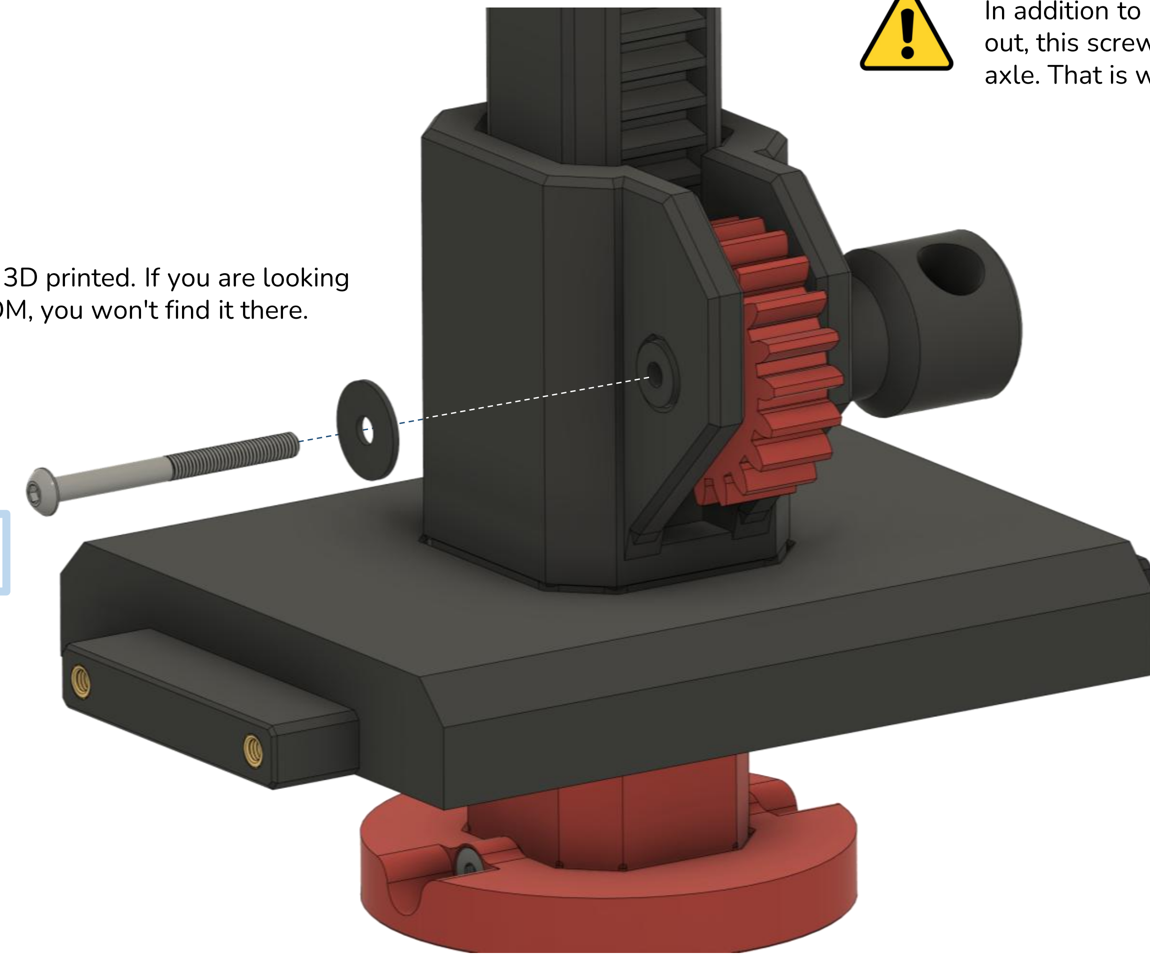
MECHANICS

TIP:

In addition to preventing the axle from falling out, this screw also serves to strengthen the axle. That is why this screw is so long.

TIP:

The washer is 3D printed. If you are looking for it in the BOM, you won't find it there.



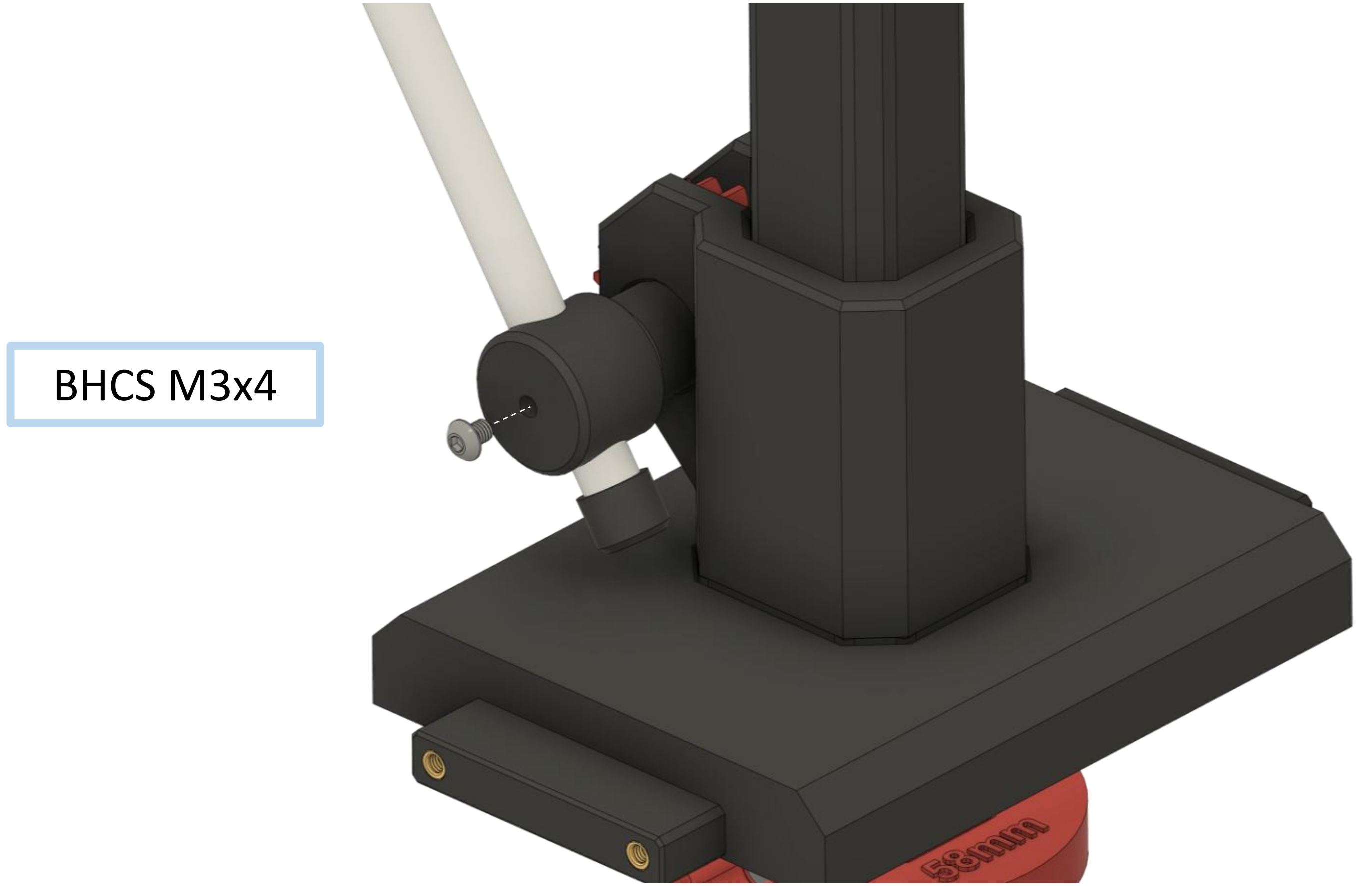
BHCS M3x35

MECHANICS

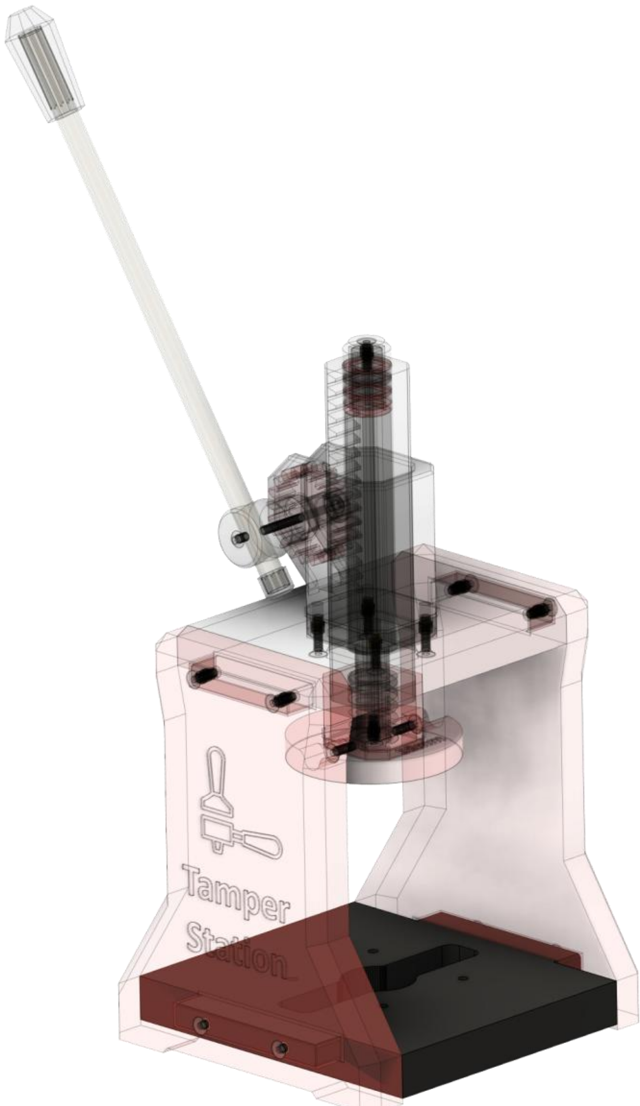
INSERT ROD

Insert the rod and then attach the end caps. The end caps are press-fit, if they do not hold you can help with a little **superglue**. Another method is to stick a piece of electrician's tape under the pressing point until the press-fit is created.

MECHANICS



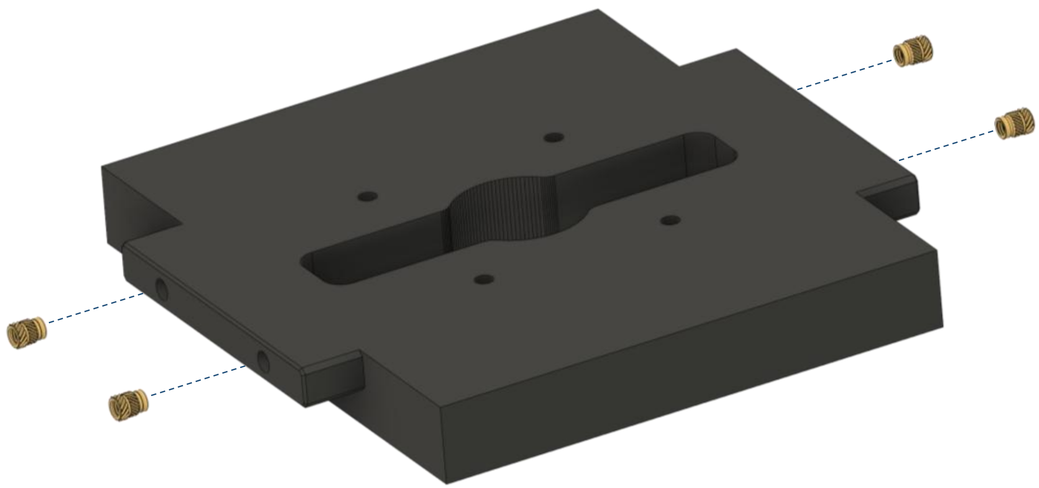
TIP:
Be careful not to overtighten the screw, as it is a screw that is screwed directly into the plastic.

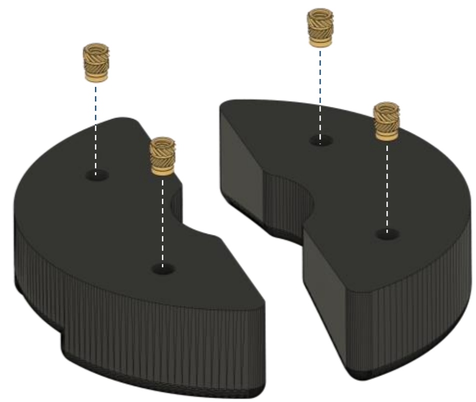


BASE BOTTOM ASSEMBLY

BASE BOTTOM

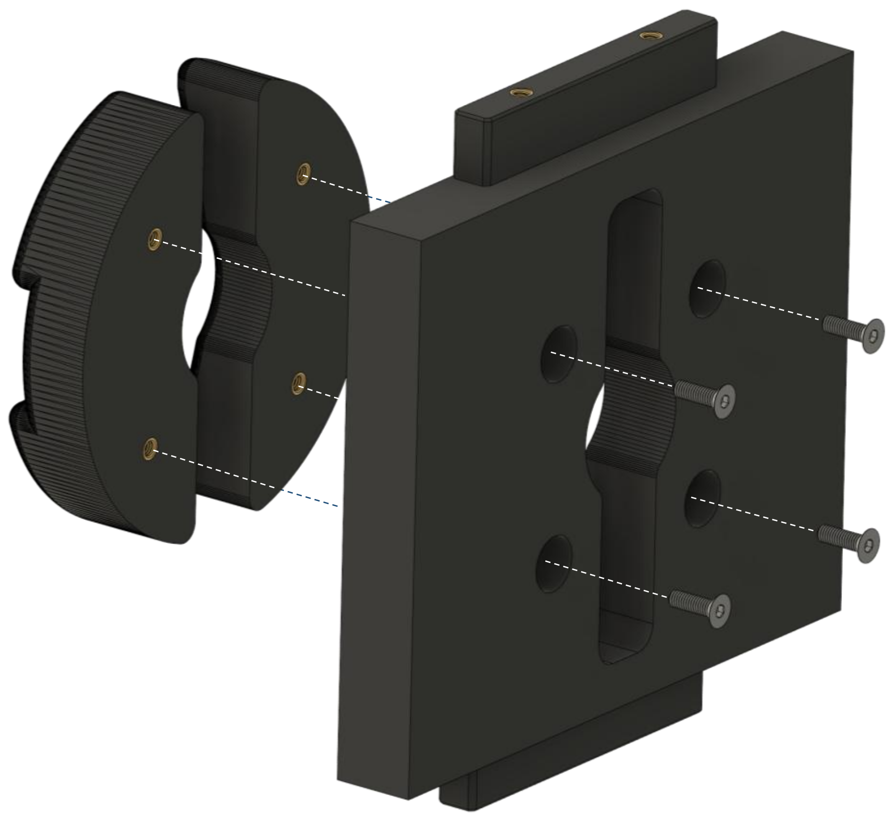
M3 Heat set insert



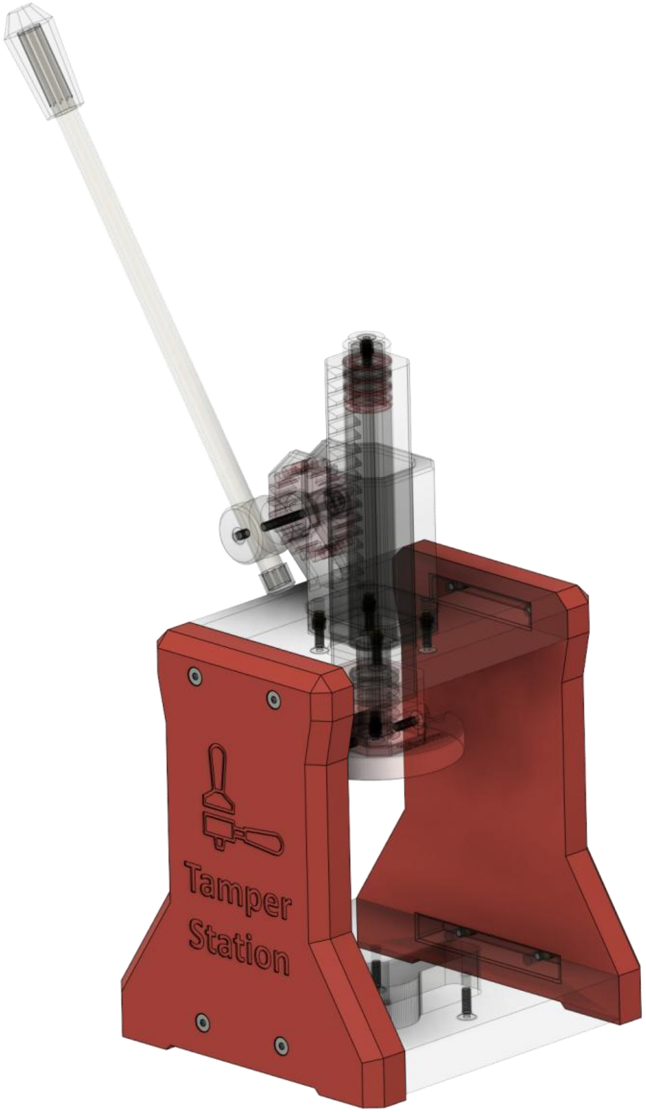


TIP:
The instructions show an example of how to assemble the bottom base holder for a 58mm GAGGIA portafilter. The actual assembly depends on the variant you choose but is always somewhat similar.

BASE BOTTOM

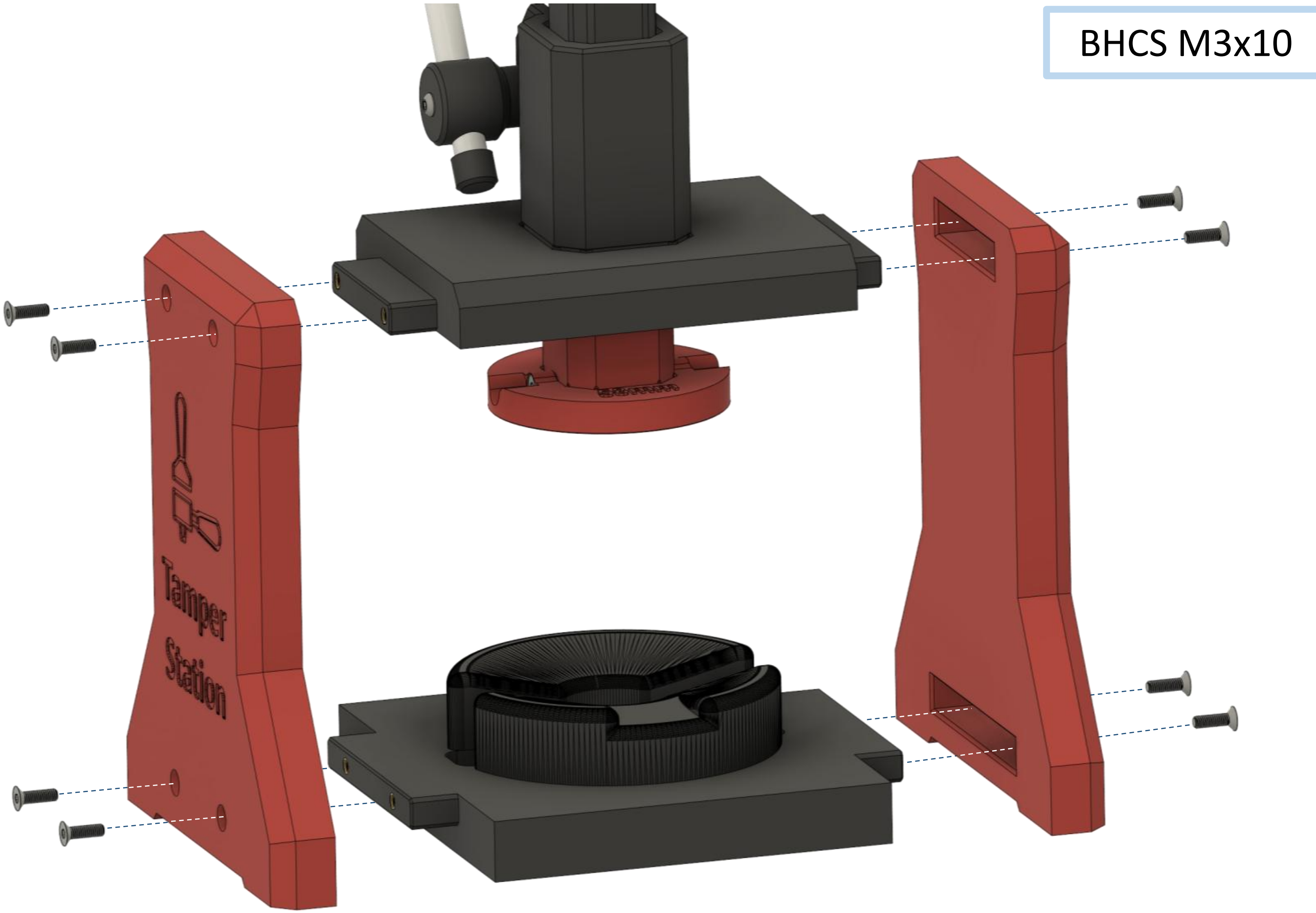


BHCS M3x10

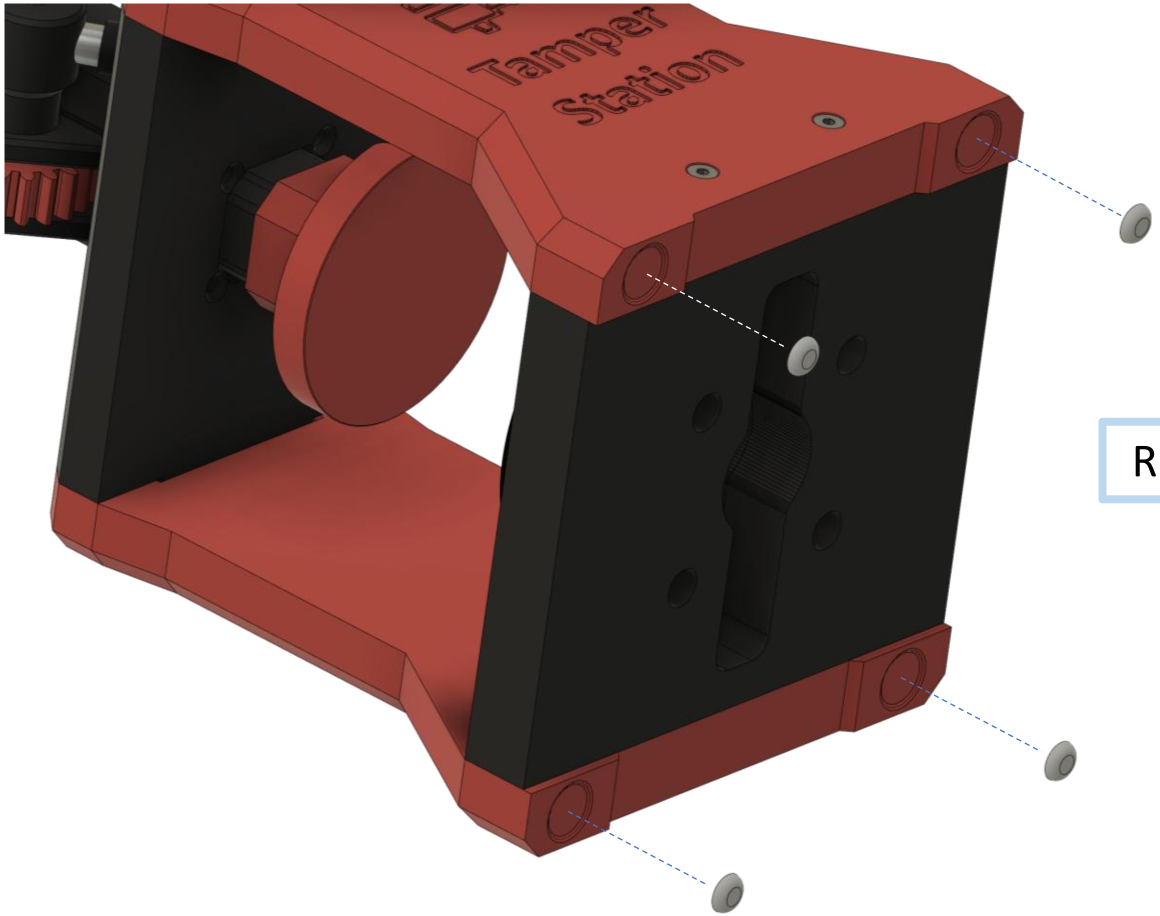


BASE SIDES ASSEMBLY

BASE SIDES



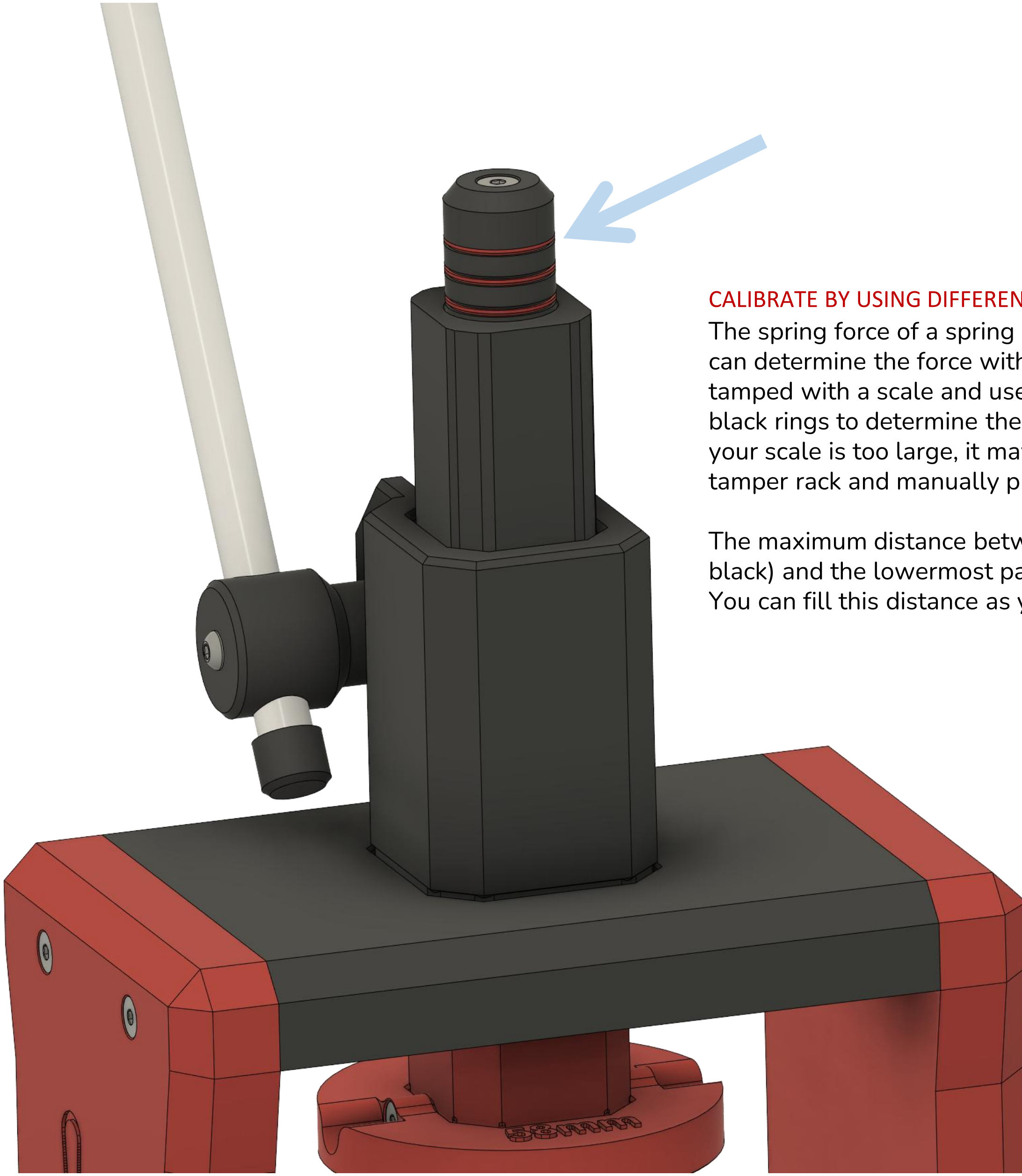
BASE SIDES



Rubber Pad 8mm

CALIBRATION

CALIBRATION



CALIBRATE BY USING DIFFERENT RING SIZES

The spring force of a spring is linear to its compression. You can determine the force with which the coffee powder is tamped with a scale and use a suitable combination of red or black rings to determine the right pressure for your setting. If your scale is too large, it may be necessary to remove the tamper rack and manually press the upper part onto a scale.

The maximum distance between the uppermost part (always black) and the lowermost part (always red) is 13-13.5 mm. You can fill this distance as you wish.

ASSEMBLY COMPLETED

TAMPER
STATION

ASSEMBLY COMPLETED! ...
Enjoy your Tamper Station.